

SurveyTalk: Voice-to-UI Mapping for In-Situ ESM via LLMs

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Abstract

The Experience Sampling Method (ESM) allows researchers to collect ecologically valid data on users' daily experiences. However, conventional mobile ESM systems rely on visual interfaces (e.g., checkboxes, text fields), which can burden users in mobile or hands-free contexts. Voice-based alternatives address this but are often limited by rigid, rule-based interactions. We present SurveyTalk, a multimodal ESM system powered by large language models (LLMs) that enables real-time semantic mapping between speech and visual survey interfaces. The system supports voice-driven selection, clarification, navigation, and response interpretation, reducing user friction and improving data quality. Our approach enhances the accessibility, adaptability, and robustness of ESM in dynamic real-world settings.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; Systems and tools for interaction design; Visualization systems and tools.

Keywords

experience sampling method, large language models, multimodal interaction, mobile survey systems, conversational interface

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1 Introduction

The Experience Sampling Method (ESM) is a widely used technique for capturing users' emotions, behaviors, and thoughts in real-world settings through real-time self-reports, offering ecologically valid and temporally precise data by minimizing recall bias [2, 11]. With the ubiquity of smartphones—owned by over 85% of adults and checked more than 50 times [1, 10] a day—mobile ESM systems have enhanced the feasibility and granularity of in-situ data collection [12]. Unlike stationary smart speakers, smartphones enable context-aware ESM prompts across users' daily lives.

Despite their benefits, mobile ESM systems often rely on touch-and typing-based interfaces, such as checkboxes and text fields, which can burden users, especially when they are on the move, visually fatigued, or unable to use their hands [12]. These constraints often lead to low compliance and high dropout rates, particularly in long-term studies. To address this, recent studies had explored voice-based ESM using smart speakers or mobile microphones [5, 6]. For example, Lim et al. [5] introduced a smart speaker-based mobile ESM that delivers context-aware prompts and supports voice or display touch input.

However, such systems remain largely rule-based, limited to recognizing predefined commands, and do not support semantically interpret natural utterances, filtering irrelevant input, or adapting the survey flow dynamically. When users give irrelevant responses or encounter dynamic branching—where later questions depend on earlier answers—they may misinterpret the question's intent. Without the ability to revisit or verify responses, this can degrade data quality or cause lost survey entries.

Meanwhile, recent work leveraging large language models (LLMs) has demonstrated their potential to enhance self-report systems by classifying or summarizing natural language input [13], or simulating user behavior to support the design of ESM protocol [4]. However, these approaches typically operate offline or in text-only settings and do not support real-time, conversational interaction

during ESM tasks. In this work, we present a SurveyTalk, a multi-modal ESM system powered by LLMs, designed to enable natural, flexible, and intelligent interactions during in-the-moment data collection. Our system bridges spoken-language input and visual survey interfaces through semantic reasoning, allowing users to interact with ESM surveys using voice in a way that directly maps to structured mobile User Interface (UI) components.

This design enables a dynamic multi-modal survey experience by fully integrating visual and speech elements. Users speak naturally, the system interprets their intent, and the UI updates in real time. Compared to visual-only ESM interfaces requiring attention and manual input (e.g., checkboxes, buttons, and text fields), our approach reduces the burden and supports hands-free use. Unlike rigid, command-based voice-only systems, our LLM-powered design enables flexible dialogue, semantic understanding, and dynamic survey flow. Participants can interact more comfortably—even while walking, multitasking, or in visually constrained settings—with real-time clarification when needed. For researchers, the system enhances compliance, captures richer contextual data, and semantically structures responses, improving the quality and scalability of in-situ ESM studies.

2 SurveyTalk: Mapping Voice-to-Visual UI in a Multi-Modal ESM System Implementation

To support seamless voice-based self-reporting, we developed SurveyTalk, a multi-modal ESM system that semantically maps spoken responses to visual UI elements in real time. The system consists of three core modules: Proactive Notification, Multi-modal Voice-to-visual UI Mapping, and Data Storage & Analytics (Figure 1).

In the **Proactive Notification** system, the system proactively encourages user participation through voice or visual alerts based on user-defined conditions (e.g., random alert enabled, number of alerts per day, alert intervals). This functionality is implemented using the *React Native Push Notification* library[7] and *Google TTS API* [3], which allows each survey item to be played by text-to-speech and responded to by voice or touch input.

In the **Multi-modal Voice-to-Visual User Interface Mapping** system, the users' voice responses are first transcribed through a speech-to-text (STT) engine [8] and then analyzed by the *GPT-4o-mini* LLM [9] in real time. The model determines the user's intent and context to update the UI state accordingly. The LLM prompt is structured into three components: a *System Instruction* that defines overall behavior, a *Decision Rule* that guides the model to classify each utterance into predefined intent categories, and a *Few-Shot Prompting* scheme to ensure response stability. This framework enables precise mapping of user utterances to UI actions. For example, when an off-topic response is detected, the system automatically classifies it as OFF-TOPIC and prompts the user with clarification such as "Could you please say that again?" to elicit a relevant response.

Finally, in the **Data Storing & Analytics** stage, only responses that are initially relevant or become relevant after a clarification prompt are considered valid. These are then stored on the backend in a structured format.

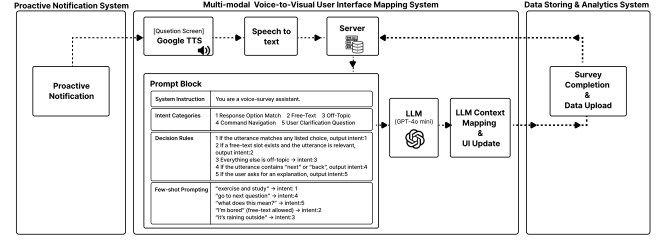


Figure 1: End-to-end SurveyTalk ESM pipeline showing three stages (1) Proactive Notification, (2) Multi-modal Voice-to-Visual UI Mapping, and (3) Data Storing & Analytics and the four dialogue sub-steps: TTS output, STT capture, LLM intent classification, and UI update

3 SurveyTalk Interaction Scenarios

We present six scenarios to illustrate the core functionalities of our proposed SurveyTalk ESM system. Each scenario demonstrates how the system detects and processes user responses across varying contexts. These examples highlight how our multi-modal ESM system flexibly adapts to a range of situations encountered during real-world use. The system dynamically interprets and responds to changes in user input, enabling reliable data collection even in complex or ambiguous response conditions. Visual representations and flow diagrams for each scenario are provided in Figure 2.

(A) Dynamic Branching Our system implements a dynamic branching mechanism in which follow-up questions are adaptively generated in real time based on the user's previous response. For example, when a user answers, "I was having a meal during a meeting," the system interprets this as two concurrent activities. Consequently, the subsequent questions are adjusted to reflect both Work/Study and Eating/Snacking contexts, allowing the participant to continue answering for each activity separately.

(B) Response Option Match by Speech The system supports direct voice input, allowing users to respond to survey questions using natural speech. For instance, when a user says, "I was in a meeting," the system employs an LLM to semantically interpret the utterance and automatically map it to the corresponding response option on the visual interface (e.g., Work/Study), checking and saving the appropriate checkbox.

(C) Free-Text If the user provides a voice response that does not match predefined survey options but is semantically relevant, the system uses the LLM to interpret it. If valid, it selects the "other" option and records the transcribed utterance as text.

(D) Off-Topic When the user's response is semantically unrelated to the survey question (i.e., off-topic), the system detects the irrelevance using the LLM and prompts the user to try again. For example, if a user responds "It's raining" to the question "Could you tell me what was most important to you during your activity?", the system identifies the mismatch and issues a voice prompt such as "I'm not sure that answers the question, could you answer again?" before restarting the speech recognition (STT) process to guide a more appropriate response.

(E) Navigation Command The system supports voice-based navigation commands. Users can control the survey flow by saying

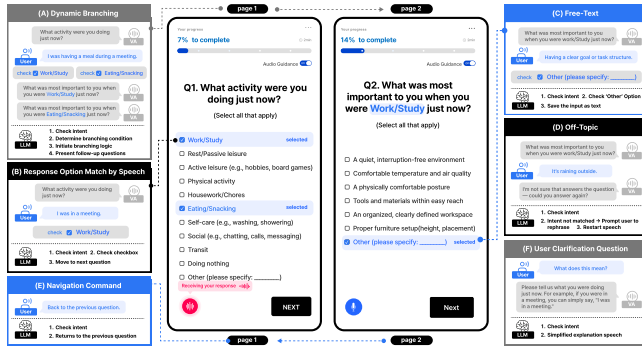


Figure 2: SurveyTalk Interaction Scenarios

commands such as “Back to the previous question,” prompting the system to transition the UI accordingly.

(F) User Clarification Question If the user expresses confusion during the response process, e.g., “I don’t understand what that means,” the system provides a simplified explanation of the question via voice or text, helping users better understand and respond.

4 Preliminary Validation

We conducted a preliminary validation of 150 utterances (30 per intent category) to evaluate the end-to-end system from speech recognition to UI mapping. The resulting classification accuracies were 93.3% for Option Match, 90.0% for Free-Text, 93.3% for Command Navigation, 96.7% for User Clarification Question, and 100.0% for Off-Topic.

5 Discussion & Conclusion

This work demonstrates how integrating LLMs into mobile ESM systems can fundamentally improve the flexibility, accessibility, and semantic understanding of in-situ self-reporting. By bridging spoken language and visual UI components in real time, our multi-modal approach enables users to interact with surveys naturally through voice, while still benefiting from the clarity and structure of visual interfaces. This design not only reduces cognitive and situational burden in mobile contexts but also empowers the system to handle ambiguous, off-topic, or unclear responses with adaptive clarification. Compared to prior rule-based voice systems, our LLM-powered interaction model offers a more intelligent, robust, and user-friendly experience.

In this study, we employed a fixed set of few-shot examples at the prototype stage. For future, longer-term work, we plan to boost performance by adopting a lightweight Retrieval-Augmented Few-Shot (RA-FS) approach that dynamically retrieves examples semantically similar to the input utterance and injects them into the prompt, thereby expanding example diversity. RA-FS addresses the limitations of fixed examples while remaining optimized for resource-constrained mobile devices, and it can eventually be extended to Retrieval-Augmented Generation (RAG) for context-aware classification and responses. Through extended field studies, we will quantitatively evaluate system usability, speech-recognition robustness, and the accuracy gains achieved by RA-FS, further solidifying the practical scalability and scholarly contribution of SurveyTalk.

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